

Hardware Components & System Assembly

Section A — Concept Application

1. Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

I would first choose a **motherboard** that supports the required CPU socket, RAM type, and storage connection. Then, I would select a **compatible CPU** that matches the motherboard socket. Next, I would choose the correct **RAM type and capacity** supported by the motherboard, and select a **SATA or M.2 storage device** supported by the board. Finally, I would check the **manuals** to confirm that the CPU, RAM, and storage are all compatible with the motherboard before purchasing.

2.Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

Points	HDD	SSD	NVMe
Speed	Slow	Fast	Very Fast
Technology	Moving disks	Flash memory	Flash memory + PCIe
Moving Parts	Yes	No	No
Connection	SATA	SATA	M.2/PCIe
Best Use	Large, low-cost storage	Office & daily work	High-performance tasks

Recommendation: **SSD** is suitable for office work because it is fast, reliable, and provides quick boot and application loading.

3. Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

The **GPU** processes graphics and helps the design software run smoothly. The **cooling system** removes heat produced by the GPU and other components during long working hours. Good cooling prevents **overheating**, which helps maintain performance and system stability.

4. Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Hardware compatibility is important because all computer components must **fit and work together correctly**.

Possible problems with incompatible components:

- The PC may **not boot**.
- Components may **not be detected**.
- Parts may **not physically fit**.
- The system may have **errors or poor performance**.
- Some components may **not work properly**.

Therefore, compatibility should be checked before assembling the PC.

5. Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

I would choose a **PSU with enough wattage** to safely power the CPU, GPU, motherboard, storage, and other components.

Safety and connection checks:-

- Turn off the PC and **disconnect the power**.
- Follow **ESD safety precautions**.
- Connect the **24-pin ATX cable** to the motherboard.
- Connect the **PCIe 6/8-pin cable** to the GPU, if required.
- Connect the **SATA power cable** to the storage device.
- Check that all connectors are **firmly connected** before switching on.

This ensures the system has suitable power and reduces the risk of connection problems before the first boot.

6. Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

I would perform these checks to confirm the workstation passes **POST (Power-On Self-Test)**:

- **Power on** the computer and check that it starts normally.
- Check that the system reaches the **BIOS/boot screen**.
- Look for any **error messages, warning lights, or beep codes**.
- Check that the main hardware is detected correctly.
- If POST fails, I would **switch off the PC and check the hardware connections**, especially RAM, CPU, power cables, and motherboard connections.
- After fixing the problem, I would **power on again and check POST**.

Section B — Practical Coding Tasks

Task 1: Component Identification & Functions

Components	Function	Make / Model or Specification
Motherboard	Connects all major computer components together.	BaseBoard Product HP86C1
CPU	Processes instructions and performs calculations.	Intel(R) Core(TM) i3-1005G1 CPU @ 1.20GHz
RAM	Temporarily stores data and programs currently in use.	4.0 GB DDR4
GPU	Processes graphics and displays images/video.	Intel (R) UHD Graphics
Storage Device	Permanently stores Windows, applications, and files.	Samsung MZVLQ512HALU-000H1, 512 GB NVMe SSD

Task 2: Motherboard Ports & Socket Identification

Part	Name	Purpose
1	CPU Socket	Holds and connects the CPU
2	RAM Slots (DIMM)	Holds RAM modules
3	PCIe Slot	Used for GPU and other expansion cards
4	M.2 Slot	Used for NVMe SSD
5	SATA Ports	Connects SATA HDD/SSD
6	24-pin Power Connector	Supplies power to the motherboard
7	CPU Power Connector	Supplies power to the CPU
8	USB Header	Connects front USB ports
9	Audio Header	Connects front-panel audio
10	Fan Header	Connects cooling fans

Task 3 : RAM & Storage Installation

Steps

1. **Power off the PC** and unplug the power cable.
2. Use **ESD precautions** and handle components carefully.
3. **Remove the RAM** by opening the RAM slot clips.
4. **Reinstall the RAM** and press it down until the clips lock.
5. **Remove the storage device** carefully.
6. Check the connection type: **SATA or NVMe**.
7. Reinstall the storage device correctly.
8. Start the computer and check that **RAM and storage are detected** in BIOS or Windows.

Task 4: Full Desktop PC Assembly

1. **Switch off and unplug** the PC. Use ESD precautions.
2. **Install the CPU** carefully into the motherboard socket.
3. **Install the RAM** into the correct RAM slots until the clips lock.
4. **Install the storage device** (SATA SSD/HDD or NVMe SSD).
5. **Install the motherboard** inside the PC case.
6. **Install the GPU** into the correct PCIe slot.
7. **Install the PSU** and connect the required power cables.
8. Check that **all cables and components are firmly connected**.
9. Connect the monitor, keyboard and power cable.
10. **Power on the PC** and check the BIOS/boot screen.
11. Confirm that the system completes **POST successfully** without error messages or warning beeps.

Section C — Mini Capstone Project

1. Three hardware components to check first when PC fails to boot

Component	Why check it?
PSU	If power is not supplied correctly, the PC will not start.
RAM	Loose or incorrectly installed RAM can stop POST and cause boot failure.
Motherboard/CPU	Wrong connection or CPU compatibility problems can prevent the system from starting.

2. Motherboard–CPU Compatibility

The motherboard and CPU must have a **compatible socket and chipset**. If the CPU does not match the motherboard socket, the system may not boot or recognize the processor. I would check the motherboard model, CPU model, socket type, and manufacturer compatibility list.

3. RAM and Storage Verification

- Check that RAM is fully inserted and the clips are locked.
- Open BIOS/UEFI and confirm the correct RAM capacity is detected.
- Check that the SSD/HDD is connected properly.
- Confirm the storage device appears in BIOS/UEFI.
- During POST, check for error messages or warning beeps.

4. Hardware Installation Checklist

1. Check all components for compatibility before installation.
2. Use ESD protection and safely handle components.
3. Make sure RAM, CPU, storage and cables are properly seated.
4. Check PSU power connections before starting.
5. Power on and confirm successful POST.

5. Why SSD instead of HDD?

SSD is better for a modern office workstation because it provides faster boot time, faster application loading and better overall responsiveness. SSDs also have no moving parts, so they are quieter and generally more resistant to physical shock than HDDs.

Section D — AI-Augmented Learning

Step 1 — Build with AI

Create a step-by-step troubleshooting checklist for a junior technician to diagnose a desktop PC that fails to boot immediately after assembly. Put the steps in correct priority order and include motherboard, RAM, storage and PSU checks. Apply the checklist to the ABC Solutions boot-failure scenario.

AI Original Checklist

1. Check the PSU and power cable.
2. Check motherboard power connections.
3. Check RAM installation.
4. Check CPU and motherboard connections.
5. Check storage connection.
6. Power on the PC and check POST.
7. Check BIOS/UEFI for detected hardware.
8. If the problem continues, check individual components.

Step 2 — Test & Debug Without AI

1. **Power OFF and unplug the PC.**
2. **Use ESD precautions** before touching components.
3. Check PSU switch, power cable and PSU connections.
4. Check motherboard **24-pin and CPU power connectors**.
5. Check CPU installation and motherboard compatibility.
6. Remove and reinstall RAM; confirm it is properly locked.
7. Check SSD/HDD connection and installation.

8. Power ON and check **POST/BIOS**.
9. Confirm RAM and storage are detected.
10. If the problem remains, test components individually.
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